

PROJECT 5 — BI DASHBOARD MINI PROJECT

Sector Research & Data Gathering

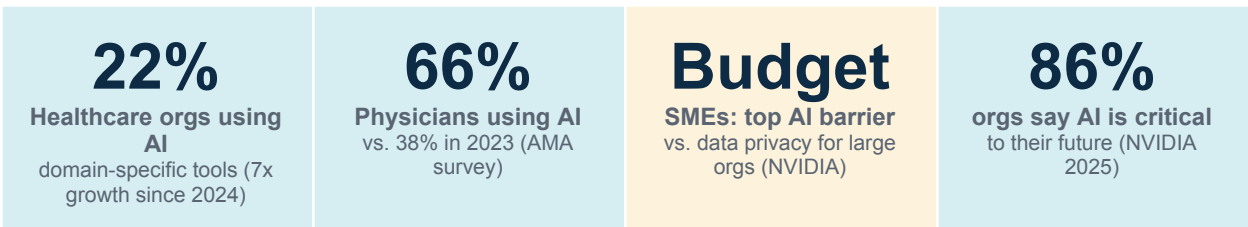
Healthcare (SME, 51–250 employees) | AI Consulting Discovery

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Executive Summary

The healthcare sector is undergoing rapid AI-led transformation. For an SME operating in this space, the window for early adoption is open — but so are the risks around regulation, data privacy, and change management. This document synthesises current sector trends, identifies the primary AI opportunities and risks relevant to a 51–250 person healthcare organisation, and documents the datasets selected to support the BI dashboard and monitoring deliverables.

1. Sector Snapshot: Healthcare in 2026



Healthcare AI adoption accelerated sharply through 2024–2025. A clear inflection point emerged in early January 2025, when outpatient and ambulatory care organisations showed the steepest growth in AI uptake — from 4.6% (2023) to 8.7% (2025) (JAMA, 2025). Despite this, healthcare still lags behind finance, education, and tech in absolute adoption rates, creating both a risk (falling behind) and an opportunity (first-mover advantage within the sector) for SMEs willing to invest.

Key insight for Chleo: Healthcare is moving fast. An SME that builds AI-ready operations now will be significantly better positioned than peers still evaluating in 12 months.

2. Key Sector Trends

2.1 Clinical Documentation Automation

Ambient AI scribing (AI that listens to patient consultations and auto-generates clinical notes) is the fastest-adopted AI use case in healthcare. It was the only use case achieving 100% adoption activity across surveyed US health systems, with 53% reporting high success (JAMIA, 2025). Tools like Abridge and Ambience have rapidly gained market share even from large incumbents. For an SME, lightweight LLM-based note summarisation is a realistic, high-impact starting point.

2.2 Operational Efficiency and Scheduling

Administrative burden is a persistent pain point — the AHA estimates healthcare administrative costs are skyrocketing, with physicians spending insufficient time on direct care due to documentation load. No-show prediction and intelligent scheduling are two AI applications with direct ROI for ambulatory and outpatient SMEs: reducing missed appointments by even 10–15% delivers measurable revenue recovery.

2.3 Generative AI and RAG for Patient Engagement

Retrieval-Augmented Generation (RAG) chatbots are gaining traction for patient intake, FAQ handling, and referral triage. These systems allow organisations to deploy AI-driven communication that is grounded in the organisation's own documentation — reducing hallucination risk and increasing explainability, which directly addresses concerns about AI transparency.

2.4 Governance and the Shadow AI Problem

In 2025, shadow AI surged across healthcare organisations as staff independently adopted GenAI tools to handle burnout and documentation pressure (Wolters Kluwer, 2025). 2026 is being called the year of governance — health system leadership is now playing catch-up. For an SME like Chleo's, this means the opportunity is to build governed AI from the start, rather than inheriting technical debt from ungoverned adoption.

2.5 Regulatory Intensification

The regulatory environment is tightening. In January 2025, HHS OCR proposed the first major HIPAA Security Rule update in 20 years, introducing stricter AI-specific requirements. In Europe, GDPR mandates transparency and explainability for automated decision-making. State-level legislation (Connecticut SB 10, Montana HB 556, Maryland HB 1240) is restricting AI from replacing clinical judgement. Compliance-by-design is now a business requirement, not a nice-to-have.

3. Key Sector Challenges

Challenge	Detail & SME Implication
Budget constraints	For SMEs under 1,000 employees, budget is the #1 AI adoption barrier (NVIDIA, 2025). Upfront tooling and integration costs must be weighed against measurable ROI.
Data quality and fragmentation	Healthcare data lives in siloed EHR systems, spreadsheets, and paper forms. AI models require clean, structured inputs — significant data prep effort is typical before any model can be deployed.
HIPAA/GDPR compliance	Any AI system touching patient data (PHI) requires a Business Associate Agreement with the AI vendor, de-identification protocols, audit logging, and breach notification readiness. Only 23% of health systems had BAAs in place for AI tools as of 2025 (PMC, 2026).
Clinician trust and adoption	The 'black box' perception is real. Staff will not trust or consistently use AI tools that cannot explain their outputs. Explainability and monitoring are non-negotiable for clinical settings.

Change management	Workflow integration requires training, champion identification, and iteration. SMEs lack dedicated change management teams, making phased rollouts essential.
Vendor lock-in risk	AI-specific healthcare vendors (Epic AI, Oracle Health AI) often bundle AI into existing contracts. SMEs need to evaluate build-vs-buy and ensure data portability.

4. Data Sources for This Project

The following publicly available datasets were identified and selected to support the dashboard, no-show prediction model, and LangSmith monitoring deliverables:

Dataset	Source, Size & Relevance
Medical Appointment No-Shows	Kaggle (joniarroba/noshowappointments) — 110,526 appointments, 14 variables including age, gender, scholarship, chronic conditions, SMS reminder, and no-show outcome. Primary dataset for dashboard and prediction use case.
Hospital Mortality Prediction	Kaggle — ICU patient records with outcome labels. Useful for demonstrating clinical risk stratification AI.
Synthetic Patient Intake Dataset	Generated using Faker (Python) — synthetic patient intake records with complaint categories, triage priority, and wait time. Used for the LangSmith monitoring evaluation set (avoids PHI entirely).
Global Health Expenditure	WHO Global Health Observatory — country-level healthcare spending data. Used for sector context visualisations in the dashboard.
Healthcare AI Market Size	Statista / Grand View Research public summaries — projected market growth data. Used in executive summary slide of dashboard.

All datasets used are either fully de-identified, synthetic, or public domain. No real patient PHI is processed at any stage of this project.

5. Research Process

The following process was used to gather and validate the sector intelligence above:

- Primary literature scan: PMC/JAMIA (peer-reviewed AI adoption studies, 2025–2026)
- Industry reports: Menlo Ventures, NVIDIA, Wolters Kluwer, Bessemer Venture Partners (2025–2026 health AI reports)
- Regulatory tracking: HHS OCR, HIPAA Journal, Foley & Lardner (HIPAA/GDPR AI compliance updates, 2025)

- Dataset discovery: Kaggle search with filters 'healthcare', 'medical appointments', sorted by usability score > 8.0
- SME-specific filter: Each insight was evaluated through the lens of a 51–250 employee organisation — excluding enterprise-only solutions and focusing on free-tier or low-cost tooling